



Home Office

Home Office Standard for Police Chemical Irritant Sprays: PAVA

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1. Management Summary

The management summary below provided the historical and current amendments that have been made to this standard.

Technical Specification – 1997

In 1997 the Centre for Applied Science and Technology (CAST) produced a technical specification to ensure that CS sprays (containing o-Chlorobenzylidene Malononitrile) supplied to the police were manufactured to a consistent standard and were designed to meet Police Operational Requirements. This has been updated a number of times, to incorporate the approved irritant pelargonic acid vanillylamide (PAVA) in 2005, with the approved solvent formulations and to reflect experience gained from operational deployment.

CAST Standard for Police Chemical Irritant Sprays: CS and PAVA (Publication Number: 23/14) – 2014

In 2014 a revision to the Standard was published to include the submission of a technical file detailing the construction of a device, independent chemical analysis and formal batch testing through the Manufacturing and Quality Testing (MQT).

Home Office Standard for Police Chemical Irritant Sprays: PAVA (Publication Number 23/14) – Version 2 – October 2024 (here on in referred to as the Standard)

This revised Standard supersedes all pre-existing specifications and standards. It includes updates that been made to provide clarity on the technical specification and the certification process required for PAVA sprays to meet Police Operational Requirements, reflecting the changes since the Standard was published in 2014.

The major revisions include:

- References to CS and CAST removed from the Standard following changes to Police Operational Requirements and the integration of CAST into Dstl in 2018. The exception being the Management Summary.
- Removal of sections 5.11 Trial to determine accuracy and handling qualities, 5.14 Familiarisation of training sprays and section A & B detailing the results tables and questionnaire for the accuracy and handling trial.
- Removal of part 2 through to part 5 within the method of section 5.10 Compatibility of spray with conducted energy devices
- Addition of requirements for the technical file now detailed in Appendix A.
- An update to the flammability of spray assessment to incorporate the review of evidence for flammability testing required by the Aerosol Dispenser Regulations (or an equivalent) for production of aerosols.
- An update to the compatibility of sprays with conductive energy devices and impact pressure of spray to solid surfaces assessments to reflect changes to equipment.
- An additional associated document has been developed to provide process and guidance on Home Office Certification for Police Chemical Irritant Sprays: PAVA.

This Standard is fully supported by the National Police Chiefs' Council Self Defence, Arrest and Restraint (SDAR) Portfolio.

2. Introduction

This Standard sets out the requirements for a PAVA spray that must be met, in order to gain Home Office Certification. Where applicable the test details the facilities required, methods for measurements together with the requirements to be met and detail for reporting.

The Standard details the technical file evidence that is required to be provided and agreed prior to testing of the PAVA spray against requirements in this Standard.

The standard outlines specific test requirements that would need to be complied with, and number of devices that would be provided which will depend on if the PAVA spray is:

- (a) A new device undergoing certification for the first time;
- (b) An existing certified device, with no changes/ modifications undergoing the 5 yearly Manufacturer Quality Testing (MQT);
- (c) A change/modification to an existing certified device.

The Home Office uses a Nominated Test Laboratory (NTL) to conduct assessment of end-to-end certification activities for each application submission for certification of Police chemical irritant spray product based on PAVA in line with this Standard. This primarily consist of conducting a technical file review assessment, conducting testing and providing the results to Home Office who will make a decision on issuing certification.

An associated Home Office Certification for Police Chemical Irritant Sprays: PAVA Process and Guidance Document provides guidance information to this standard including suggested templates that may be wished to be followed.

3. Scope

3.1 Overall Scope

The Standard details the requirements for a PAVA spray that must be met such that sufficient evidence is gained to meet Police Operational Requirements.

The Standard does not outline regulatory requirements for the manufacture of aerosol devices and their component parts. It is the responsibility of the manufacturers and suppliers of the sprays to ensure that these requirements are met. This will not form part of any Home Office (HO) evaluation. Guidance on these regulations may be sought from the British Aerosol Manufacturing Association.

This Standard covers PAVA sprays and inert training devices.

3.2 Definition

The Police chemical irritant spray product based on PAVA is also referred to as 'device' or 'PAVA spray' throughout this standard.

4. Certification Overview

4.1 Certification

The Home Office uses a NTL to conduct assessment of end-to-end certification activities for each application submission for certification of Police chemical irritant spray product based on PAVA in line with this Standard. This primarily consist of conducting a technical file review assessment, conducting testing and providing the results to Home Office who will make a decision on issuing certification.

This Standard details the requirements and test methods for PAVA sprays that must be met in order to achieve Home Office certification. The specific test requirements that would need to be complied with, and number of devices that would need to be provided will depend on if the PAVA spray is a new device, has a change/modification to an existing certified device or undergoing a MQT.

A new device undergoing certification for the first time

All new police PAVA sprays must meet all requirements in the standard and if compliant a new certificate is issued and will expire five years after the certification issue date.

A change/modification to an existing certified device

Certification issued is only valid for the particular design of device, manufactured in accordance with the technical file submitted as part of the original certification. If there is a design change (modification) to a device at any point during the five year validity of certification it must be resubmitted for recertification with an updated technical file. The device must meet all requirements in the standard and will be retested to ensure sustained compliance with the Standard. If compliant a new certificate will be issued and will expire five years after the certification issue date.

An existing certified device, with no changes/ modifications undergoing the 5 yearly MQT

Once Home Office certification is issued, manufacturers are responsible for ensuring continued performance of their devices throughout production, via their quality assurance process. The Home Office requires a MQT programme. This is a formal batch testing method ensuring the continued quality of a device once it has met the requirements of the Home Office certification. MQT for previously certified devices with no modifications shall be conducted on a five-yearly basis from the original certification date. It requires manufacturers to submit a number of devices for MQT and if compliant a certification will be issued and will expire five years after the certification issue date.

The Home Office reserves the right to undertake additional assessments of devices at any time, should it be found necessary.

Any applicants that are entering the certification process must be aware that all data required for certification, including the technical file and results of testing, will be retained by Home Office and could be shared within government, police forces and Home Office Technical Partner. Applicants must allow NTL access to their application and technical files, solely for the purpose of the NTL involvement in conducting independent assessment of PAVA Sprays.

4.2 General Requirements for Submission of PAVA Sprays

Prior to submission of devices for compliance testing, applicants must provide a technical file (in English) detailing the information listed in **Appendix A**. A review of the information provided in technical file will be completed and agreed prior to the commencement of any testing. This is a requirement for certification and must be completed.

4.3 Tests and Quantities

The tests required for a new or change/ modification to an existing certified device are shown in Table 1 along with the number of devices required for each test. The tests required for a MQT are shown in Table 2 along with the number of devices required for each test. In all cases the devices for testing shall be selected randomly from a production batch where a batch is defined as a set of devices filled from a single volume of solution.

Table 1: Tests and quantities of devices required for full certification testing

Section	Test	No. Devices Required (upright only)	No. Devices Required (upright & inverted)
5.2	Inspection of Labelling	1 live	1 live
5.2	Resistance to Crushing	3 live	3 live
5.5	Drop Testing	14 live	17 live
5.6	Determination of Discharge Rate	3 live	6 live
5.7	Determination of Impact Pressure of Spray to Solid Surface	3 live	6 live
5.8	Performance at Extreme Temperatures	15 live	21 live
5.9	Assessment of Spray Flammability ¹	3 live	6 live
5.10	Compatibility of Spray with Conducted Energy Devices	2 live	2 live
5.11	Chemical Analysis ²	5 live	5 live
5.12	Inert Devices	3 inert	6 inert
Device spares		6 live 5 inert	8 live 5 inert

¹ To be conducted if the solution is classified as flammable from the Aerosol Dispenser Regulation testing (ADR 2009) or equivalent. Information should be provided in the technical file.

² In addition to the devices, in accordance with Section 5.11: Chemical analysis a minimum of 0.7 g pure PAVA should be provided.

Including spares, the total number of devices required for full certification testing is 55 live and 8 inert. If it is specified that the device will work in an inverted orientation the total number of devices required (including spares) is 75 live and 11 inert devices.

Table 2: Tests and quantities of devices required for MQT

Section	Test	No. Devices Required (upright only)	No. Devices Required (upright & inverted)
5.3	Resistance to Crushing	3 live	3 live
5.5	Drop Testing ³	3 live	6 live
5.6	Determination of Discharge Rate	3 live	6 live
5.8	Performance at Extreme Temperatures ⁴	6 live	12 live
5.11	Chemical Analysis ^{2, 5}	5 live	5 live
Device spares		5 live 5 Inert	5 live 5 inert

Including spares, the total number of devices required for MQT is 25 live and 5 inert. If it is specified that the device will work in an inverted orientation the total number of devices required (including spares) is 37 devices and 5 inert devices.

4.4 Certification Requirements

4.4.1 Technical File

A review of the information provided in technical file (**Appendix A**) will be completed and agreed prior to the commencement of any testing. This is a requirement for certification and must be completed.

4.4.2 Testing

The PAVA sprays is required to pass all of the tests (depending on the type of certification (full certification or MQT)) specified within this Standard to receive certification. Requirements particular to each assessment are described in the relevant section.

If a device fails to meet the requirements at any stage during assessment, testing will be discontinued and the applicant will be alerted.

³ Positions 1, 3 and 5 only

⁴ Spray pattern assessment only

⁵ Chemical analysis for MQT is required. Applicants can use their batch data for the MQT and must submit full analytical results for the batch of devices submitted for the MQT assessment. The analytical methods used shall be the same (or equivalent) as those presented in the original technical file used for initial certification. If applicants choose not to submit batch data as part of MQT, they shall use the Home Office NTL. See Section 5.11 Chemical analysis of PAVA solution

The applicant will be given one opportunity to investigate the issue and resubmit their device for testing, ensuring the technical file is updated to reflect the resolution of the issue. Prior to resubmission of the device the applicant is required to demonstrate why the issue occurred and provide the NTL and Home Office with an update on the remedial action taken to rectify the problem.

4.4.2.1 Generic Testing Requirements

Where this section is referenced in the requirements section for a specific test, the following generic requirements shall be met by all devices:

- shall be discharged when the actuator is activated (e.g. button depressed);
- shall stop discharging when the actuator is de-activated (e.g. button released);
- shall not be discharged as a result of any action other than intended discharge;
- shall not be damaged so that the device cannot physically be discharged without remedial action (e.g. actuator falls off or significant damage to the cap);
- shall not leak.

4.5 Certification Outcome

Upon successful completion of the assessment applicants will be issued with a certificate of compliance, which will expire five years after the certification date. If during MQT previously certified devices are found to fail to meet requirements, certification will be suspended until the manufacturer/supplier resolves the issues.

Home Office will have the final decision on issuing a certificate.

Once a product is certified Home Office shall be informed if:

- (a) Issues are identified by the applicant, relating to the assurance of PAVA sprays during quality control inspections.
- (b) There is any design change to a device, in which case devices are to be resubmitted to be reassessed to ensure sustained compliance with the Standard.

5. Testing and Requirements

5.1 Capacity

5.1.1 Standard devices

Standard-sized devices shall contain a minimum of 30 ml PAVA solution, there is no maximum capacity for PAVA spray.

5.1.2 Concealable devices

Devices containing between 10 ml and 30 ml shall be considered as “Concealable”. These shall be assessed according to the same tests as standard devices; however, some requirements differ. The differing requirements for concealable devices are specifically stated where relevant.

5.2 Inspection of Labelling

Note: It is the responsibility of the applicant to ensure that regulatory labelling information is included on each device⁶. This will not form part of the Home Office assessment.

5.2.1 Facilities required

- Number of devices as outlined in Table 1
- Container with a capacity great enough to accommodate the entire contents of the device e.g. glass beaker
- Glass pipette
- Stop watch

5.2.2 Method

The device shall be inspected to ensure that it includes the specific information for police PAVA sprays as detailed in Section 5.2.3

The entire contents of one device shall be decanted into a suitable container. The label and base shall be covered in the decanted PAVA solution using a pipette, left for 2 minutes and the results of the test recorded.

5.2.3 Requirements

Standard devices:

The following required information shall be included on each device:

- Nature of intended use, e.g. “PAVA Spray, for Police Use Only”
- Supplier’s or manufacturer’s name, address and telephone number
- Name of main ingredients, e.g. “contains PAVA in ethanol, water and monopropylene glycol”
- A number or code uniquely identifying each individual device

⁶ Guidance may be sought from the British Aerosol Manufacturing Association

- Date of manufacture
- Date at which shelf life expires

Compliance with any part of any standard shall not be indicated on the device.

The required information shall remain on the device and be readable following a 2 minute exposure of the label and base, to the PAVA solution contained within the device.

Concealable devices:

The following required information shall be included on each device:

- A number or code uniquely identifying each individual device
- Date at which shelf life expires

Compliance with any part of any standard shall not be indicated on the device and shall not specify that they are for police use.

The required information shall remain on the device and be readable following a 2 minute exposure of the label and base, to the PAVA solution contained within the device.

5.2.4 Reporting

The information of results is required to be recorded. This should include as minimum:

- Date of test
- Manufacturer
- Type of device
- Device capacity
- Device number
- Yes / No for required information on device
- Yes / No for readability post test

5.3 Resistance to Crushing

Note: an assessment of spray pattern (Section 5.4) follows this test.

5.3.1 Facilities required

- Lever device as illustrated in Figure 1 (the distance between the centres of the pivot and attachment holes must be 1500 mm)
- Table clamps
- Weight suitable for suspension from lever (see Figure 1 and calculation below)
- Impermeable bag (e.g. polythene) with capacity at least 20 times that of the device
- Balance capable of measuring up to 20 kg ± 0.1 kg
- Number of devices as outlined in either Table 1 or Table 2
- Conditioning cabinet maintained at 20°C ± 1°C
- Test laboratory should maintain temperature at 20°C ± 5°C

Calculation of weight (W) for suspension from lever:

$$W = 20 - \left(\frac{wl}{1500} \right)$$

Where: w is the weight of the arm in kg

l is the distance in mm between the centre of gravity of the arm and the pivot (see Figure 1)

Before testing, the devices shall be conditioned for 30 minutes at 20°C ± 1°C. Each test on each device should be completed in the shortest time possible after removal from conditioning; ideally within five minutes and up to a maximum of ten minutes.

5.3.2 Method

This test simulates the effect of a person weighing 100 kg stepping on the device. The lever device shall be clamped firmly to a bench using table clamps. The device shall be enclosed in the bag, the air excluded and the bag sealed. The weight shall be suspended from the end of the lever.

The device shall be placed under the lever with the foam rubber pad aligned with the bottom of the aerosol container portion of the device (see Figure 1 – Position A). The lever shall be lowered onto the device and the position of the device checked and, if necessary, adjusted. It may be found necessary to secure the device (e.g. with adhesive tape) to stop it slipping out from under the lever. The weight shall then be lowered gently until it is freely suspended from the lever and left for one minute. The device shall be removed from the underneath lever and checked for any leakage or damage.

The test shall be repeated with the foam rubber pad aligned with the cap of the device (Figure 1– Position B).

After the crush test is completed, each device shall be tested as described in Section 5.4 “Assessment of spray pattern”. The test shall be repeated for all devices.

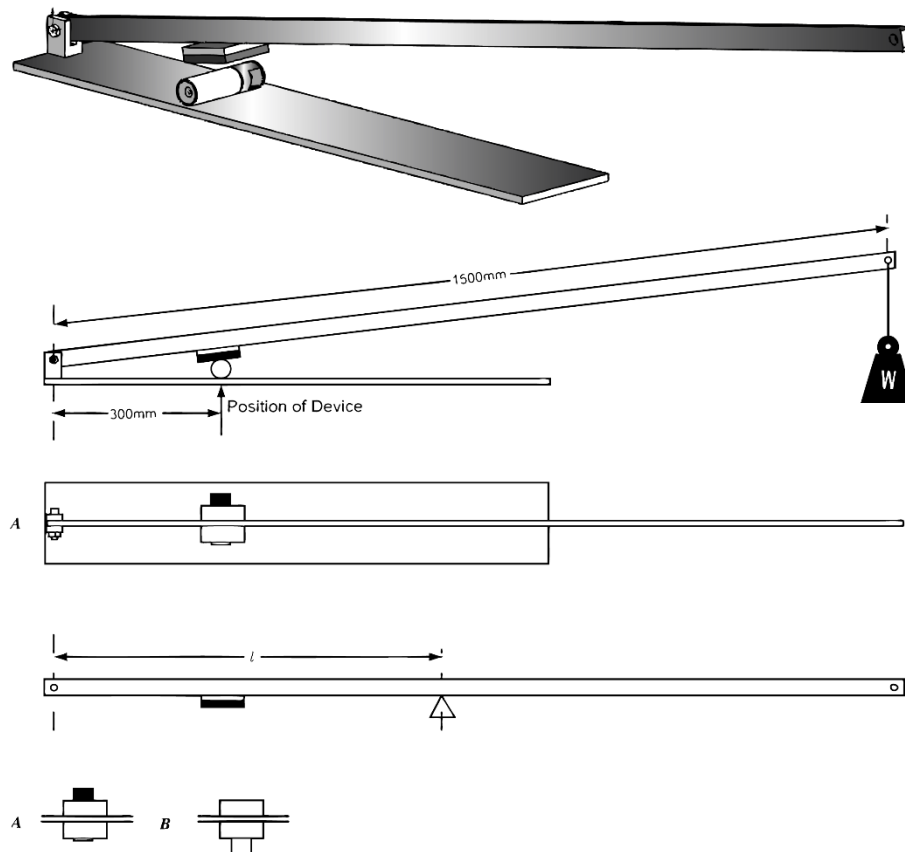


Figure 1 - Crush test rig

Suggested crush pad dimensions: 75 mm x 75 mm, 10 mm thick.

Suggested crush pad material: Nitrile/PVC closed cell foam. Compression deflection 63-91 kPa.

5.3.3 Requirements

All devices shall satisfy the generic testing requirements specified in Section 4.4.2.1 and assessment of spray pattern requirements specified in Section 5.4.3.

5.3.4 Reporting

The information of results is required to be recorded. This should include as minimum:

- Date and time of test.
- Manufacturer.
- Type of device.
- Device capacity.
- Conditioning temperature.
- Laboratory temperature.
- Device number.
- Pass/ Fail for each device for spray pattern requirements section 5.4.3.
- Pass/ Fail for each device for generic testing requirements section 4.4.2.1.

5.4 Assessment of Spray Pattern

Note: this assessment will be conducted following the tests for Resistance to Crushing (Section 5.3), Drop Testing (Section 5.5) and Performance at Extreme Temperatures (Section 5.8).

5.4.1 Facilities required

- Tape measure
- Conditioning cabinet maintained at $20^{\circ}\text{C} \pm 1^{\circ}\text{C}$
- 36 sheets of paper, at least A3 sized
- One transparent sheet (e.g. tracing paper or acetate) with a 150 mm diameter circle drawn on it
- Test laboratory should maintain temperature at $20^{\circ}\text{C} \pm 5^{\circ}\text{C}$

Before testing, the devices shall be conditioned for 30 minutes at $20^{\circ}\text{C} \pm 1^{\circ}\text{C}$. Each test on each device should be completed in the shortest time possible after removal from conditioning; ideally within five minutes and up to a maximum of ten minutes.

5.4.2 Method

The spray pattern shall be determined by spraying a half-second burst at a sheet of paper at a range of two metres. The actuator button shall be fully depressed. Most organic solvents will mark the paper. The spray pattern shall be assessed to ascertain if it falls substantially (80%) within a 150 mm diameter circle, by overlaying a transparent sheet with such a circle drawn on it. The test shall be carried out with a full device and repeated to assess a total of 12 sprays volume in the device permitting. The test shall be repeated for all three devices.

Concealable devices shall be assessed using the same method; however, the range shall be reduced to one metre. The test shall be repeated until the device becomes empty. The test shall be repeated for all three devices.

5.4.3 Requirements

The generic requirements detailed in Section 4.4.2.1 shall be met for all devices, in addition to meeting the requirements specified below.

Standard devices: 80% of each spray from each device shall fall inside a 150 mm target when discharged from a distance of 2 metres. The first six half-second sprays for each device tested shall satisfy this requirement; the subsequent six sprays should be assessed to give an indication of continued performance.

Concealable devices: 80% of each spray from each device shall fall inside a 150 mm target when discharged from a distance of 1 metre. The first four half-second sprays shall satisfy this requirement; further sprays should be assessed to give an indication of continued performance if the volume contained in the device allows.

The examples of spray patterns in Figure 2 may be used to help determine whether the spray falls substantially within the circle.

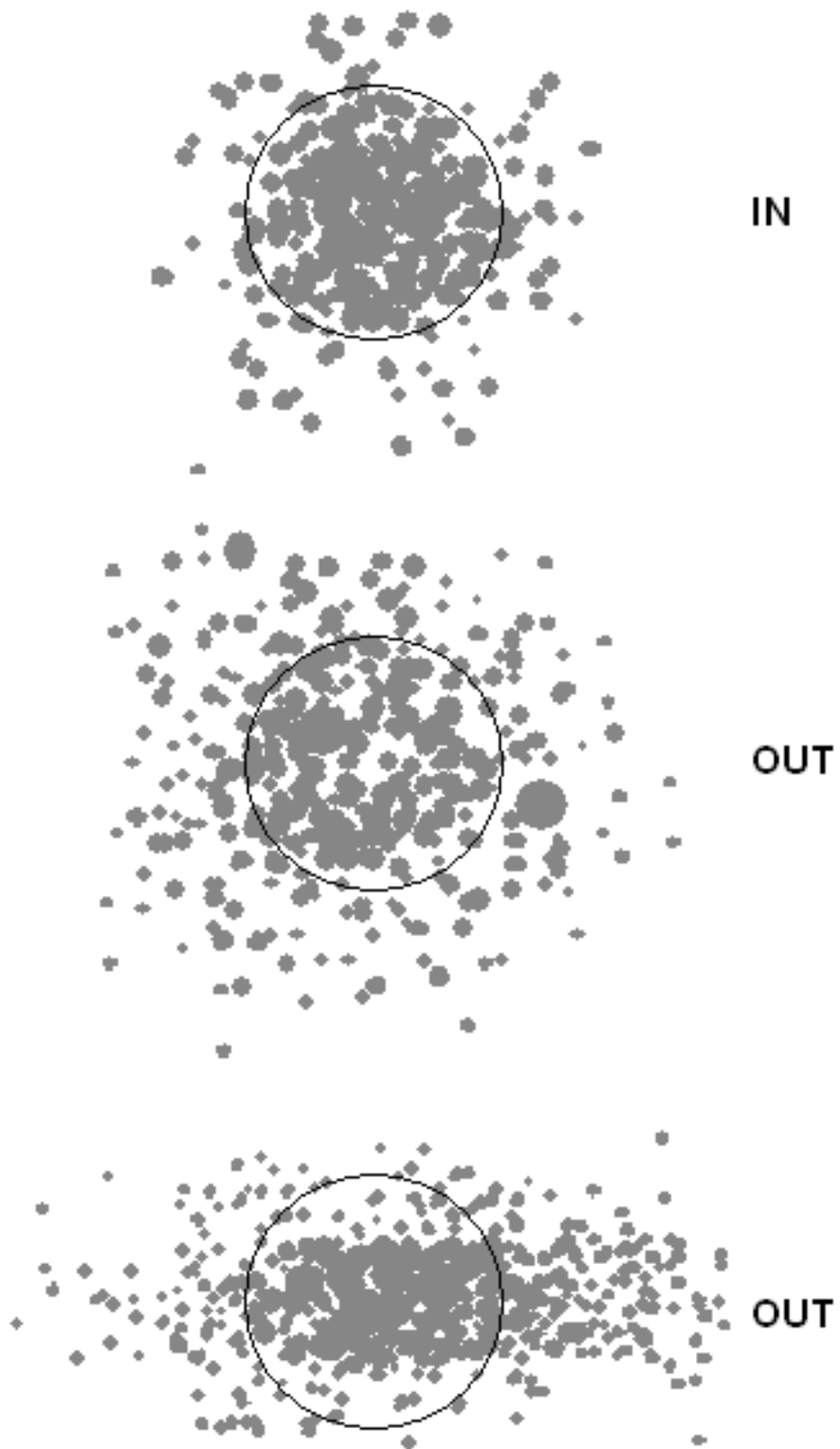


Figure 2 - Spray pattern examples.

5.5 Drop Testing

Note: an assessment of spray pattern (Section 5.4) follows this test.

5.5.1 Facilities required

- Conditioning cabinet maintained at $20^{\circ}\text{C} \pm 1^{\circ}\text{C}$
- Steel plate of at least 6 mm thickness and at least 300 mm x 300 mm
- Tape measure
- Suitable stand and clamp from which device may be suspended
- Number of devices as outlined in either Table 1 or Table 2
- Test laboratory should maintain temperature at $20^{\circ}\text{C} \pm 5^{\circ}\text{C}$

Before testing, the devices shall be conditioned for 30 minutes at $20^{\circ}\text{C} \pm 1^{\circ}\text{C}$. Each test on each device should be completed in the shortest time possible after removal from conditioning; ideally within five minutes and up to a maximum of ten minutes.

5.5.2 Method

The device shall be suspended at one of the specified angles indicated in Figure 3. The centre of the device shall be $2\text{ m} \pm 5\text{ cm}$ above the approximate centre of the steel plate, which is laid flat. The device shall be dropped by releasing the clamp. The test shall be repeated using a different device at each specified angle until all 14 specified angles have been completed. After the drop test is completed, each device shall be tested as described in Section 5.4 “Assessment of Spray Pattern.”

If it is specified by the manufacturer/supplier that the device will work in an inverted orientation a further three devices shall be tested by dropping from positions 1, 3 and 5 (Figure 3). This shall be followed by the “Assessment of Spray Pattern” (Section 5.4); however, during this assessment the device shall be mounted in an inverted position.

5.5.3 Requirements

All devices shall satisfy the generic testing requirements specified in Section 4.4.2.1 and assessment of spray pattern requirements specified in Section 5.4.3.

5.5.4 Reporting

The information of results is required to be recorded. This should include as minimum:

- Date and time of test.
- Manufacturer.
- Type of device.
- Device capacity.
- Conditioning temperature.
- Laboratory temperature.
- Orientation.
- Device number.
- Pass/ Fail for each device for spray pattern requirements section 5.4.3.
- Pass/ Fail for each device for generic testing requirements section 4.4.2.1.

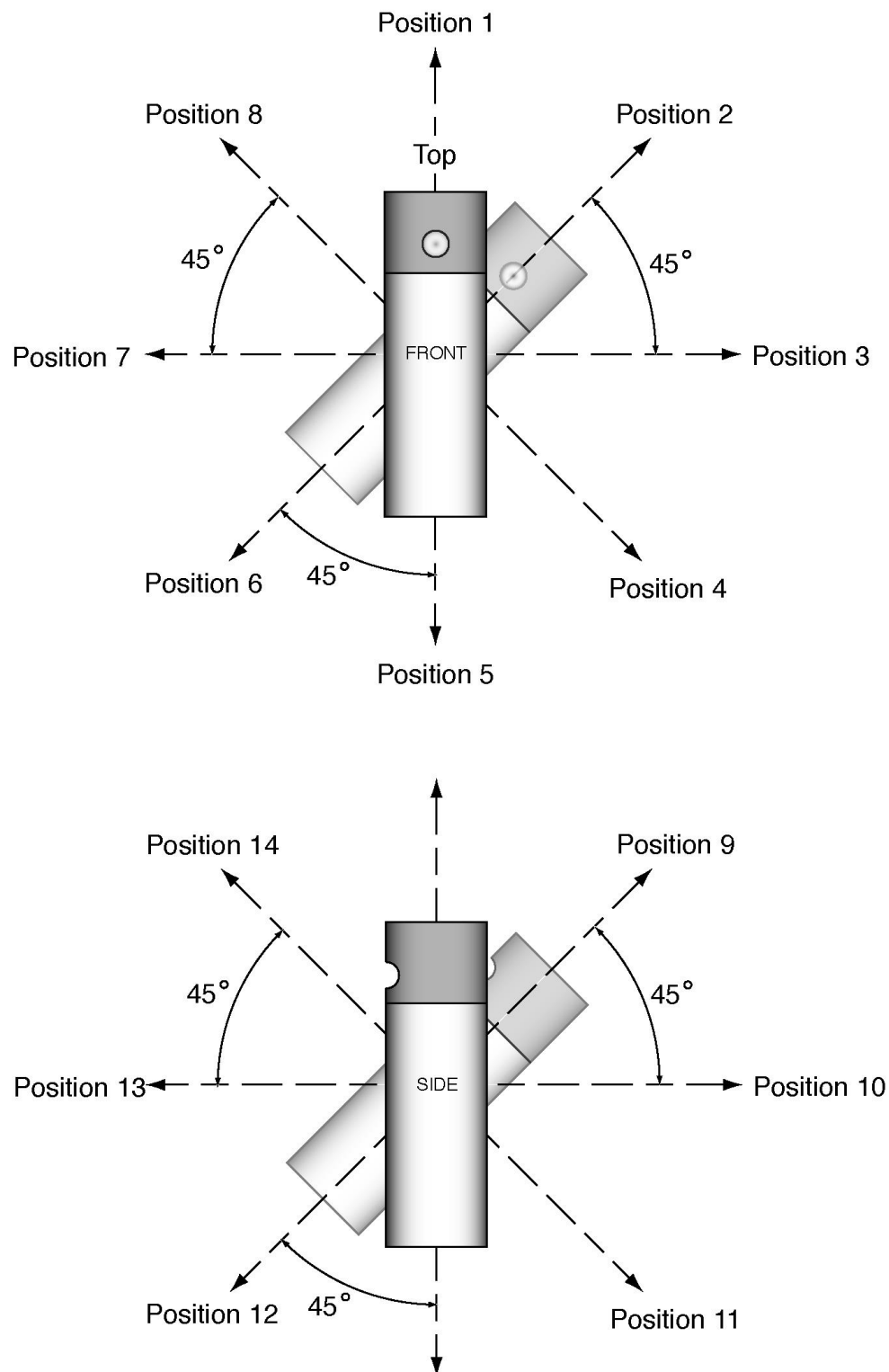


Figure 3 - Specified angles for drop test.

5.6 Determination of Discharge Rate

5.6.1 Facilities required

- A conditioning cabinet maintained at 20°C ± 1°C
- A balance accurate to 0.01 g
- Timing apparatus accurate to 0.01 seconds. Suggested timing devices: high-speed camera or speaker cone/microphone and oscilloscope
- Number of devices as outlined in either Table 1 or Table 2
- Test laboratory should maintain temperature at 20°C ± 5°C

Before testing, the devices shall be conditioned for 30 minutes at 20°C ± 1°C. Each test on each device should be completed in the shortest time possible after removal from conditioning; ideally within five minutes and up to a maximum of ten minutes.

5.6.2 Method

Standard devices: Each device shall be weighed three times and individual results shall be recorded. The arithmetic mean shall be calculated and reported. The device shall be discharged for approximately 0.5 seconds (the exact length of time of discharge shall be measured using a suitable timing apparatus). The actuator button shall be fully depressed. Any liquid which adheres to the device shall be removed before the device is reweighed three times. The individual results shall be recorded and arithmetic mean shall be calculated and reported. This process shall be repeated so that a total of 12 discharges are assessed.

Concealable devices: The method for standard devices shall be followed and repeated until the device becomes empty. It is not necessary to assess 12 discharges.

The test shall be repeated for all devices.

If it is specified by the manufacturer/supplier that the device will work in an inverted orientation then this test shall be repeated in this orientation.

5.6.3 Calculation of discharge rate

The weight of each discharge (arithmetic mean) and the length of each discharge in seconds shall be recorded for each device. The discharge rate in ml/s shall be calculated at the end of the test.

The discharge rate is calculated using the difference in weight before and after each burst and the density of the liquid contained in the device, according to the following formula:

$$R = \frac{W_1 - W_2}{td}$$

Where:

R is the discharge rate in ml/s

w1 is the weight before the burst in g

w2 is the weight after the burst in g

t is the length of time of discharge in s

d is the density of the solvent in g/ml

5.6.4 Definitions for requirements

A_{min} – absolute minimum

A_{max} – absolute maximum

P – Permitted range

D – Average (arithmetic mean) discharge rate for the first six half-second discharges from an individual device

$\bar{D}$ – Average (arithmetic mean) discharge rate for the first six half-second discharges across all devices

5.6.5 Requirements

The generic requirements detailed in Section 4.4.2.1 shall be met for all devices, in addition to meeting the requirements specified below.

For any single discharge from any of the devices tested the discharge rate shall not be less than the absolute minimum (A_{min}) or greater than the absolute maximum (A_{max}).

The average (arithmetic mean) discharge rate for the first six half-second discharges across all three devices tested ($\bar{D}$) shall not exceed the figures stated in Table 3.

The permitted range (P) of results shall be calculated from $\bar{D}$ for the first six half-second discharges across all three devices:

$$Minimum = \bar{D} - 2$$

$$Maximum = \bar{D} + 2$$

The lower limit of P shall not be less than A_{min} and the upper limit of P shall not be greater than A_{max} . For example for a PAVA device if $\bar{D} = 11.5$, P would be 9.5. to 13.0.

The average (arithmetic mean) discharge rate for first six half-second discharges for each individual device (D) must be within 10% of $\bar{D}$.

Once calculated, the values for $\bar{D}$, D and P will be set for the device being tested within a set condition.

Table 3 - Discharge rate requirements

	A_{min}	A_{max}	P	$\bar{D}$	D
PAVA	2.5 ml/sec	13.0 ml/sec	$\bar{D} \pm 2$ ml/sec	No greater than absolute maximum	$\bar{D} \pm 10\%$

* Arithmetic rounding rules will be applied where relevant

5.6.6 Discharge rate worked example

An example demonstrating how to apply the requirements is shown in this section. Table 4 shows the calculated discharge rate per spray of three devices, as well as figures for D , $\bar{D}$ and P .

Table 4 - Worked example (PAVA spray)

Spray number	Device #1 Discharge rate (ml/s)	Device #2 Discharge rate (ml/s)	Device #3 Discharge rate (ml/s)
1	7.68	8.02	7.05
2	7.54	7.91	6.86
3	7.39	7.68	6.57
4	7.01	7.30	6.42
5	6.84	7.05	6.29
6	6.59	6.92	6.07
7	6.40	6.73	5.87
8	6.28	6.70	5.69
9	6.13	6.52	5.52
10	5.96	6.27	5.39
11	5.87	5.93	5.21
12	5.51	5.23	4.96
D	7.2	7.5	6.5
$\bar{D}$	7.1		
P	5.1 - 9.1		

This would be a PASS result as:

$A_{\min}$ is less than 6.07

$A_{\max}$ is greater than 8.02

All values for discharges 1 – 6 are within the limits P

D is within 10% of $\bar{D}$ (required to be between 6.36 and 7.78)

5.6.7 Reporting

The information of results is required to be recorded. This should include as minimum:

- Date and time of test.
- Manufacturer.
- Type of device.

- Device capacity.
- Conditioning temperature.
- Laboratory temperature.
- Orientation.
- Device number.
- For each device being tested:
 - Weight of device before testing (conducted 3 times) together with its average.
 - For each spray discharge
 - Spray duration
 - Weight for device (conducted 3 times) together with its average.
 - R - Discharge rate (ml/s)
- The value of D across each device – stating if pass / failed requirement.
- The value of $\bar{D}$ stating if it had pass / failed the requirement.
- Range of P stating if it have pass / failed the requirement.
- The value of $A_{\min}$ stating if it have pass / failed the requirement.
- The value of $A_{\max}$ stating if it have pass / failed the requirement.
- Pass/ Fail for each device for generic testing requirements section 4.4.2.1.

5.7 Determination of Impact Pressure of Spray to Solid Surfaces

5.7.1 Facilities required

- Number of devices as outlined in Table 1
- Conditioning cabinet maintained at $20^{\circ}\text{C} \pm 1^{\circ}\text{C}$
- Pressure sensing equipment (described in **Appendix B**)
- Tape measure
- Test laboratory should maintain temperature at $20^{\circ}\text{C} \pm 5^{\circ}\text{C}$

Before testing, the devices shall be conditioned for 30 minutes at $20^{\circ}\text{C} \pm 1^{\circ}\text{C}$. Each test on each device should be completed in the shortest time possible after removal from conditioning; ideally within five minutes and up to a maximum of ten minutes.

5.7.2 Method

The test equipment shall be set up so that the nozzle of the device is 15 cm from the pressure sensing plate and the spray hits the sensor square on. The device shall be discharged for one second with the actuator button fully depressed and the highest pressure measured over the entire duration of the spray shall be recorded. This shall be repeated until the device is empty. The test shall be repeated for all devices.

If it is specified by the manufacturer/supplier that the device will work in an inverted orientation then this test shall be repeated in that position.

5.7.3 Requirements

The generic requirements detailed in Section 4.4.2.1 shall be met for all devices, in addition to meeting the requirements specified below:

A device will also be deemed to have failed if the measured pressure, for any one second burst, exceeds the maximum pressure of 40 kPa. All devices shall pass this test.

5.7.4 Reporting

The information of results is required to be recorded. This should include as minimum:

- Date and time of test.
- Manufacturer.
- Type of device.
- Device capacity.
- Conditioning temperature.
- Laboratory temperature.
- Orientation.
- Device number.
- Pass/ Fail to in meeting determination of impact pressure of spray to solid surfaces of requirements 5.7.3.
- Pass/ Fail for each device for generic testing requirements section 4.4.2.1.

5.8 Performance at Extreme Temperatures

5.8.1 Facilities required

- Number of devices as outlined in either Table 1 or Table 2
- Conditioning cabinet maintained at $50^{\circ}\text{C} \pm 1^{\circ}\text{C}$
- Freezer cabinet maintained at $-20^{\circ}\text{C} \pm 1^{\circ}\text{C}$
- Protective gloves suitable for handling items at -20°C and 50°C
- Test laboratory should maintain temperature at $20^{\circ}\text{C} \pm 5^{\circ}\text{C}$

Nine devices shall be initially conditioned at $50^{\circ}\text{C} \pm 1^{\circ}\text{C}$ for 24 hours and six spray devices shall be initially conditioned at $-20^{\circ}\text{C} \pm 1^{\circ}\text{C}$ for 24 hours. After conditioning at the extremes of temperature the devices shall be removed from the cabinet and further conditioned for a period of 30 minutes at $20^{\circ}\text{C} \pm 1^{\circ}\text{C}$. Each test on each device should then be completed in the shortest time possible; ideally within five minutes and up to a maximum of ten minutes.

5.8.2 Method

Six devices shall be tested as described in Section 5.4 “Assessment of Spray Pattern” using the initial conditioned devices (three at -20°C and three at $+50^{\circ}\text{C}$). Note after initial conditioning at the extreme of temperature a further conditioning period of 30 minutes at $20^{\circ}\text{C} \pm 1^{\circ}\text{C}$ shall be conducted only once.

Six devices shall be tested as described in Section 5.6 “Determination of Discharge Rate” using the initial conditioned devices (three at -20°C and three at $+50^{\circ}\text{C}$). Note after initial conditioning at the extreme of temperature a further conditioning period of 30 minutes at $20^{\circ}\text{C} \pm 1^{\circ}\text{C}$ shall be conducted only once.

Three devices shall be tested as described in Section 5.7 “Determination of Impact Pressure of Spray to Solid Surfaces” using the initial conditioned devices (three at $+50^{\circ}\text{C}$). Note after initial conditioning at the extreme of temperature a further conditioning period of 30 minutes at $20^{\circ}\text{C} \pm 1^{\circ}\text{C}$ shall be conducted only once.

If it is specified by the manufacturer/supplier that the device will work in an inverted orientation then the test described in Section 5.4 shall be repeated in that position. It is not necessary to repeat tests described in Sections 5.6 or 5.7 in the inverted position.

5.8.3 Requirements

The generic requirements detailed in Section 4.4.2.1 shall be met for all devices, in addition to meeting the requirements specified below:

For devices tested as described in Section 5.4 “Assessment of Spray Pattern” using the initial conditioned devices at extreme temperature requirement of Section 5.4.3 must be met.

For devices tested as described in Section 5.6 “Determination of Discharge Rate” using the initial conditioned devices at extreme temperature requirement of section 5.6.5 must be met.

For devices tested as described in Section 5.7 “Determination of Impact Pressure of Spray to Solid Surfaces” using the initial conditioned devices at extreme temperature requirement of section 5.7.3 must be met.

5.8.4 Reporting

The information of results is required to be recorded.

For devices tested as described in Section 5.4 “Assessment of Spray Pattern” using the initial conditioned devices at extreme temperature the minimum information that should be recorded is:

- Date and time of test.
- Manufacturer.
- Type of device.
- Device capacity.
- Initial conditioning temperature.
- Laboratory temperature.
- Orientation.
- Device number.
- Pass/ Fail for each device for spray pattern requirements section 5.4.3.
- Pass/ Fail for each device for generic testing requirements section 4.4.2.1.

For devices tested as described in Section 5.6 “Determination of Discharge Rate” using the initial conditioned devices at extreme temperature the minimum information that should be recorded is outlined in Section 5.6.7, together with clearly outlining the initial conditioning temperature of the device.

For devices tested as described in Section 5.7 “Determination of Impact Pressure of Spray to Solid Surfaces” using the initial conditioned devices at extreme temperature the minimum information that should be recorded is outlined Section 5.7.4, together with clearly outlining the initial conditioning temperature of the device.

5.9 Assessment of Spray Flammability

Note: This test is only to be conducted if the device is classed as flammable, from the regulation testing required for aerosol production (ADR 2009) or equivalent. Evidence shall be provided in the submitted technical file.

5.9.1 Facilities required

- 150 mm rule
- Three candles
- A conditioning cabinet maintained at $20^{\circ}\text{C} \pm 1^{\circ}\text{C}$
- Number of devices as outlined in Table 1
- Draft-free area
- CO2 fire extinguisher
- Test laboratory should maintain temperature at $20^{\circ}\text{C} \pm 5^{\circ}\text{C}$

Before testing, the devices shall be conditioned for 30 minutes at $20^{\circ}\text{C} \pm 1^{\circ}\text{C}$. Each test on each device should be completed in the shortest time possible after removal from conditioning; ideally within five minutes and up to a maximum of ten minutes.

5.9.2 Method

Two candles of the same height shall be mounted in a holder perpendicular to the spray path so that the spray will travel through both candle flames. A third candle may be used in order to relight these candles in case of accidental extinguishment by the spray.

The device shall be positioned 150 mm from the candles positioned together. The candles may need support in order to have the flame at the correct vertical height relative to the device. The relative height of the device shall be arranged so that the lower edge of the spray passes through the top third of the candle flames. The device shall be discharged for at least two seconds with the actuator button fully depressed. It shall be noted if the flame extends back to the device.

This shall be repeated using the same device; however, the spray shall be removed from the candle flames to establish whether it continues to burn. If the spray continues to burn, the flame shall be recorded as self-sustaining.

The test shall be repeated until the device is empty. For a result to be considered valid, the spray shall not extinguish the candle flame during the assessment. The test shall be repeated for all three devices.

If it is specified by the manufacturer/supplier that the device will work in an inverted orientation then this test shall be repeated in that position.

5.9.3 Requirements

The generic requirements detailed in Section 4.4.2.1 shall be met for all devices. In addition there shall be **no** flame extension towards the device and the flame shall **not** be self-sustaining.

5.9.4 Reporting

The information of results is required to be recorded. This should include as minimum:

- Date and time of test.
- Manufacturer.
- Type of device.
- Device capacity.
- Conditioning temperature.
- Laboratory temperature.
- Orientation.
- Device number.
- Pass/ Fail for each device meeting requirements Section 5.9.3.
- Pass/ Fail for each device for generic testing requirements Section 4.4.2.1

5.10 Compatibility of Spray with Conducted Energy Devices

The flammability of the sprays shall be assessed when subjected to high-voltage, low-energy sparks. This will be used to monitor the effects of conducted energy devices on PAVA sprays. No pass/fail criteria are set for this test; however, the results shall be made available to the manufacturer and police forces.

5.10.1 Facilities required

- Spark generator – 50 kV, maximum 0.0036 Amps, 19 Hz
- Electrically conductive tape
- Mannequin or form heated to $30^{\circ}\text{C} \pm 1^{\circ}\text{C}$
- Material – 100% cotton ($0.17 \pm 0.05 \text{ kg/m}^2$)
- Conditioning cabinet held at $20^{\circ}\text{C} \pm 1^{\circ}\text{C}$
- Number of devices as outlined in Table 1
- Stop watch
- Draft-free area
- CO2 fire extinguisher
- Test laboratory should maintain temperature at $20^{\circ}\text{C} \pm 5^{\circ}\text{C}$

Before testing, the devices shall be conditioned for 30 minutes at $20^{\circ}\text{C} \pm 1^{\circ}\text{C}$. Each test on each device should be completed in the shortest time possible after removal from conditioning; ideally within five minutes and up to a maximum of ten minutes.

5.10.2 Method

The test shall be conducted in a draft-free area which can be ventilated after each test.

The mannequin torso or form shall be heated to a constant temperature of $30^{\circ}\text{C} \pm 1^{\circ}\text{C}$. Material shall be attached, allowing it to fall naturally to simulate air gaps under clothing as a t-shirt would over the body. One electrode from the spark generator shall be positioned on the outer side of the material with a 1 cm gap, with the second electrode the inner side of the material. See Figure 4 for recommended equipment set-up.

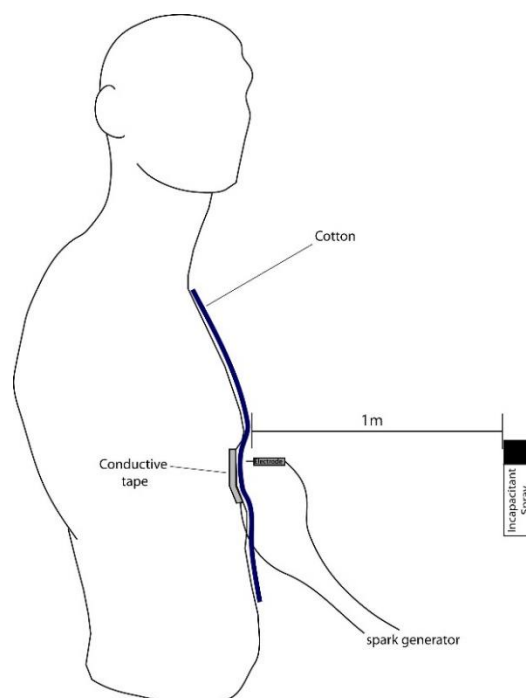


Figure 4 - Equipment set-up.

One device shall be assessed in accordance with each of Parts 1a and Part 1b.

Part 1a

From a distance of one metre two half-second bursts shall be discharged directly onto the material where the electrodes are positioned. The actuator button shall be fully depressed. After 30 seconds a pulsing spark of five seconds duration shall be generated. If no flames are observed the spark generation shall be repeated, with a five-second duration, 30 seconds after the first test. If no flames are observed during this assessment Part 1b shall be completed.

Part 1b (conduct only if spray is shown to be non-flammable in Part 1a)

Part 1a shall be repeated; with a new material and new device however, the entire contents of the device shall be discharged onto the material prior to spark generation.

5.10.3 Requirements

There are no pass/fail criteria for this assessment; the results will be provided for information. However, the device shall satisfy the general requirements detailed in section 4.4.2.1 and will be deemed to have failed if any of these requirements are not met.

5.10.4 Reporting

The information of results is required to be recorded. This should include as minimum:

- Date and time of test.
- Manufacturer.
- Type of device.
- Device capacity.
- Conditioning temperature.
- Laboratory temperature.
- Device number.
- Yes/ No for each test if flames or smouldering of the material are observed for part 1a and part 1b (if carried out).
- Pass/ Fail for each device for generic testing requirements section 4.4.2.1

5.11 Chemical Analysis of PAVA Solutions

Five devices are required for this assessment. In addition applicants shall submit a minimum of 0.7 g pure PAVA for analysis.

The five devices will be selected from the batch by the NTL for chemical analysis. The NTL will decant 20 ml solution from each of the five devices into separate glass vials and securely seal the containers. These solutions, along with the pure PAVA, will be assessed using a High Performance Liquid Chromatography with UV detector at 254nm or an equivalent analytical laboratory technique in accordance with methods by NTL.

5.11.1 Requirements

There are a limited number of approved solutions for PAVA sprays for use by UK police. These solutions are described in Table 5. If an applicant has an alternative solution based on PAVA with a different solvent they must contact the Home Office to discuss the process for approval, as this falls outside of the scope of this Standard.

The device will be deemed to have failed this requirement if the analytical results fall outside the specifications detailed below.

Table 5 - PAVA requirements

Requirement	PAVA
Irritant Agent	Pelagonic Acid Vanillyl Amide
Concentration	0.3% PAVA weight/volume
Maximum weight of irritant permitted per device	No specified limit
Approved solvents	50% aqueous ethanol Monopropylene glycol, ethanol, isopropyl alcohol and water as per relevant COT statement
Propellant	Shall be non-flammable and non-toxic. The propellant currently used is nitrogen

Purity of irritant agent in solid phase

The purity of the PAVA must be determined in the solid phase and must be at least 96.8% by weight.

Overall purity of the liquid contents of the device after filling and pressurising

For a PAVA device substances other than the solvents and PAVA must form less than 1% wt/vol of the solvents and PAVA combined.

Concentration of the liquid contents of the device after filling and pressurising

For a PAVA device no measured concentration shall be above 0.335% wt/vol or below 0.265% wt/vol. The mean of the measured concentrations shall be $0.3 \pm 0.03\%$ wt/vol.

5.11.2 Comparison of results with manufacturer/supplier data

Applicants are required to submit details of the methods they use to analyse the chemical contents of their devices and their results as part of the technical file for new submissions.

If applicants intend to use their batch data for the MQT they must submit full analytical results for the batch of devices submitted for the MQT assessment. The analytical methods used shall be the same (or equivalent) as those presented in the original technical file used for initial certification.

If applicants choose not to submit batch data as part of MQT, they shall use the Home Office NTL.

5.11.3 Reporting

The information of results is required to be recorded. This should include as minimum:

- Name and address of testing laboratory and location
- Date of receipt of samples
- Date and time of test
- Test method (including deviations)
- Unique reference number for samples that are traceable back to the manufacturer and device number
- Analytical results and the Pass/Fail for each solution from a device and pure PAVA against requirements in section 5.11.1
- Evidence of report authorisation
- Clear identification when results are from external providers

5.12 Inert Devices

Devices for use in training environments that contain only water shall be submitted.

5.12.1 Requirements

The generic requirements detailed in Section 4.4.2.1 shall be met for all devices, in addition to meeting the requirements specified below:

Determination of Impact Pressure of Sprays to a Solid Surface set out in Section 5.7.3 must be met.

5.12.1 Reporting

The information of results is required to be recorded. This is detailed in Section 5.7.4.

6. References

- 1) Specification for CS Sprays for Operational Police Use, T.G. Sheldon & Dr. J. Tan, 1997, PSDB Publication 2/97
- 2) Specification for CS Sprays for Operational Police Use: 2002 Update, T. Donnelly & M. Macfarlane, PSDB Publication 7/02
- 3) HOSDB Standard for CS and PAVA Sprays for Operational Police Use, J. Crockett, HOSDB Publication 36/05
- 4) HOSDB Standard for CS and PAVA Sprays for Operational Police Use, Revision 1, S. Croft, HOSDB Publication 38/08
- 5) Comparison of CS and PAVA Sprays: Operational and Toxicological Aspects, G. Smith, M. Macfarlane & J. Crockett, PSDB Publication 88/04
- 6) Statement on 2-chlorobenzylidene malononitrile and CS spray, Committee on Toxicity of Chemicals in Food, Consumer Products and the Environment, COT Statement: COT/1999/06, COM Statement: COM/98/S2, COC Statement: COC/98/S4, September 1999
- 7) COT statement on the use of PAVA (Nonivamide) as an incapacitant spray, Committee on Toxicity of Chemicals in Food, Consumer Products and the Environment, April 2002
- 8) Statement on the use of pava (nonivamide) as an incapacitant spray, Committee on Toxicity of Chemicals in Food, Consumer Products and the Environment, COT Statement 2004/06, November 2004
- 9) Statement: Use of PAVA (Nonivamide) as an Incapacitant Spray: Reformulation of Captor, Committee on Toxicity of Chemicals in Food, Consumer Products and the Environment, COT Statement 2007/05, July 2007
- 10) COT statement on TW1000, a new formulation of PAVA. Committee on Toxicity of Chemicals in Food and Consumer Products and the environment. COT statement 2018/03 February 2018.

Appendix A – Technical File

Applicants shall provide the below information required to be provided as part of the technical file. This has been split in to two parts. Part A - General Application Information Requirements and Part B - Technical File Details Requirements.

Part A: General Application Information Requirements

On submission of a completed Part A an applicant declaration must be signed and dated to confirm that:

- The information is correct to the best of their knowledge.
- They are aware that all data required for certification, including the technical file and results of testing, will be retained by Home Office and could be shared within government, police forces and Home Office Technical Partner. They will allow the Nominated Test Laboratory (NTL) access to application and technical files, solely for the purpose of the NTL involvement in conducting independent assessment of PAVA Sprays.

Ref	General Application Information Requirements
A1	Name of UK supplier & manufacturer
A2	Full name of product
A3	Type of product
A4	Purpose of submission: New device, modification to an existing certified device or no modification to device and approaching 5 years since device last certified.
A5	Summary of function of device
A6	Summary of changes made to device since last certified. <i>(Not required to be completed for new device)</i>
A7	Brief outline the chemical formulation of solution, confirming if this is already approved solution as indicated in Home Office standard for Police chemical Irritant Sprays: PAVA.
A8	Describe the Policing requirement, including any current and potential contractual obligation for this device or modification, and if there is policing sponsor.

Part B: Technical File Detail Requirements.

For each of the technical file detail requirements below the applicant must provide evidence to support the requirements this should be done as a commentary and/or referring to the technical file document where the information is held. If evidence to the technical file document is referred to, the full document name and page number of the information must be provided, together with the full technical document provided as attachments as part of the Part B submission.

Against each requirement, for modifications of currently certified devices details of any changes compared to the original certified device shall be clearly highlighted and explained (it is acceptable to complete only the requirements relevant to the change marking 'no change' where needed).

On submission of a completed Part B with full technical document provided as attachments, an Applicant declaration must be signed and dated to confirm that:

- The PAVA product is produced in accordance with all relevant legislation.
- All design of PAVA Spray has been finalised and devices available for the tests.
- The device will operate reliably and consistently throughout its warranted lifetime.
- The information is correct to the best of their knowledge.
- They are aware that all data required for certification, including the technical file and results of testing, will be retained by Home Office and could be shared within government, police forces and Home Office Technical Partner. They must allow the Nominated Test Laboratory (NTL) access to application and technical files, solely for the purpose of the NTL involvement in conducting independent assessment of PAVA Sprays.
- They are authorised to make the submission on behalf of the supplier.

Ref	Technical File Detail Requirements
B1	Evidence of UK supplier's authority to possess, purchase, acquire, manufacture, sell or transfer prohibited weapons under Section 5 Firearms Act 1968.
B2	Evidence of full breakdown of solvent formulation detailing the composition of the PAVA solution contained in the device Confirmation if this is already approved solution as indicated in Home Office standard for Police chemical Irritant Sprays: PAVA.

B3	Evidence and results from the flammability testing of aerosols in accordance with Aerosol Dispensers Regulations: 2009 No 2824 or equivalent.
B4	Evidence that the following information is included on the label: Standard devices. • Nature of intended use, e.g. "PAVA spray, for Police Use Only". • Supplier's name, address and telephone number • Name of main ingredients e.g. "contains PAVA in ethanol and water". • A number or code uniquely identifying each individual device. • Date of manufacture. • Date at which shelf life expires. Compliance with any part of the standard shall not be indicated on the device. Concealable devices. • A number or code uniquely identifying each individual device. • Date at which shelf life expires. Compliance with any part of the standard shall not be indicated on the device and shall not specify that they are for police use.
B5	Evidence of the propellant used and that it is non-flammable and non-toxic.
B6	Evidence that the purity of the PAVA in the solid phase is greater than or equal to 96.8% by weight.
B7	Evidence of the analytical methods used to analyse the chemical contents of PAVA devices for internal batch testing purposes.
B8	Evidence of the results of internal batch testing to include points below 8.1 – 8.3 B8.1 - Evidence that for a PAVA device no measured concentration is above 0.335 %wt/vol or below 0.265 %wt/vol. after filling and pressurising.

	B8.2 - Evidence that the mean of the measured concentrations is 0.3 ± 0.03 %wt/vol after filling and pressurising. B8.3 - Evidence that substances other than the solvents and PAVA is less than 1% wt/vol of the solvents and PAVA combined after filling and pressurising.
B9	The density of the solution at 20°C.
B10	Shelf-life of the device and evidence that the product is stable over the maximum period of the shelf-life.
B11	Evidence of the construction of the device including device assembly method.
B12	Details of the supplier for each component and chemical for the device.
B13	Hazard information for the PAVA Device. Hazard information must include any information relating to Control of Substances Hazardous to Health (COSHH) regulations or any safety precautions required in handling the device.
B14	Detail and evidence of working to recognised quality control systems in the design and manufacture of device
B15	Detail and evidence of suitable sampling and inspection for device. This is to include plans for internal batch testing for traceability from individual components, chemicals and testing to identify any defect.
B16	Confirm if device to be used in upright or both the upright and inverted positions.
B17	Declaration statement and evidence that the PAVA device/product is produced in accordance with all relevant legislation and to a recognised quality control standard.
B18	Evidence that the device will operate reliably and consistently throughout its warranted lifetime.

Appendix B – Pressure Sensing Equipment

Pressure sensor array, mounted on flat aluminium plate.

Minimum Specifications:

Number of elements	1024
Individual element dimensions	2 x 2 mm
Entire array dimensions	64 x 64 mm
Sensor frequency	5 Hz
Pressure range	0 – 12 psi (0 – 83 kPa)

Use:

Prior to experimentation the accuracy of the sensor will be tested. Pressure readings from the software must be accurate to within 10% of a calibrated pressure reference. If not, the system must be recalibrated prior to use.

Immediately post-experimentation, the accuracy of the sensor must be tested again. Pressure readings from the software must be accurate to within 10% of a calibrated pressure reference. If not, the system must be recalibrated and the test re-run.

Set-up:

